

Object Detection in Computer Vision

What Is Object Detection?

Object detection is a computer vision task that requires both locating and classifying objects within an image. Unlike image classification, which assigns a single label to the entire image, detection produces a set of bounding boxes, each paired with a class label and a confidence score.

A bounding box is typically described by four values: the x and y coordinates of the top-left corner, and the width and height of the box. Some formats instead use (x_min, y_min, x_max, y_max).

Two-Stage Detectors

Two-stage detectors first generate a set of region proposals (candidate boxes) and then classify each proposal. R-CNN (Regions with CNN features) was the pioneering work: it used selective search to produce ~2000 proposals per image and ran a CNN on each independently.

Fast R-CNN improved efficiency by passing the full image through a CNN once and projecting proposals onto the resulting feature map. Faster R-CNN replaced selective search with a Region Proposal Network (RPN) trained end-to-end with the detection head, enabling real-time performance.

One-Stage Detectors

One-stage detectors skip the proposal stage entirely and directly predict bounding boxes and class probabilities from the full image in a single forward pass. This makes them much faster, at a small cost in accuracy for small objects.

YOLO (You Only Look Once) divides the image into a grid; each cell predicts boxes and classes directly. SSD (Single Shot MultiBox Detector) uses feature maps at multiple scales to detect objects of different sizes.

Modern versions like YOLOv8 and RT-DETR have closed much of the accuracy gap while maintaining high speed, making one-stage detectors the default choice for many production applications.

Anchor Boxes

Many detectors use anchor boxes: pre-defined boxes of various scales and aspect ratios placed across the image. The network learns to predict offsets from these anchors rather than absolute coordinates, simplifying the regression problem.

The number and shape of anchors are important hyperparameters. K-means clustering on the training set bounding boxes is a common way to choose anchor shapes that match the dataset.

Anchor-Free Detection

Recent methods like FCOS and CenterNet avoid anchors altogether. FCOS predicts the distance from each foreground pixel to the four sides of its ground-truth box. CenterNet represents objects by their center point and regresses width, height, and other properties from there.

Anchor-free methods simplify the design and reduce sensitivity to anchor hyperparameters.

Evaluation Metrics

The primary metric is mean Average Precision (mAP). For each class, the precision-recall curve is computed by varying the confidence threshold; the area under this curve is the Average Precision (AP). mAP averages AP across all classes.

Intersection over Union (IoU) determines whether a predicted box matches a ground-truth box. A prediction is a true positive only if its IoU with a ground-truth box exceeds a threshold, typically 0.5. COCO evaluation averages mAP across IoU thresholds from 0.5 to 0.95.

Transformer-Based Detection

DETR (Detection Transformer) by Carion et al. (2020) reformulated detection as a set prediction problem using a Transformer encoder-decoder. It eliminates anchors and NMS, directly outputting a fixed-size set of (class, box) predictions.

Follow-up models like Deformable DETR and DINO address DETR's slow convergence, achieving strong results on COCO while retaining the clean end-to-end design.